

# HOMOGENEOUS COORDINATES

Computer Vision with Hüseyin Özdemir

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} A_{00} & A_{01} \\ A_{10} & A_{11} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} b_0 \\ b_1 \end{bmatrix}$$

To compute affine transformation with above equation,

One matrix-vector multiplication and one vector addition should be performed

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} A_{00} & A_{01} \\ A_{10} & A_{11} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} b_0 \\ b_1 \end{bmatrix}$$

To compute affine transformation with above equation,

One matrix-vector multiplication and one vector addition should be performed

Single matrix-vector multiplication would be more practical and more efficient

The solution is to use homogeneous coordinates

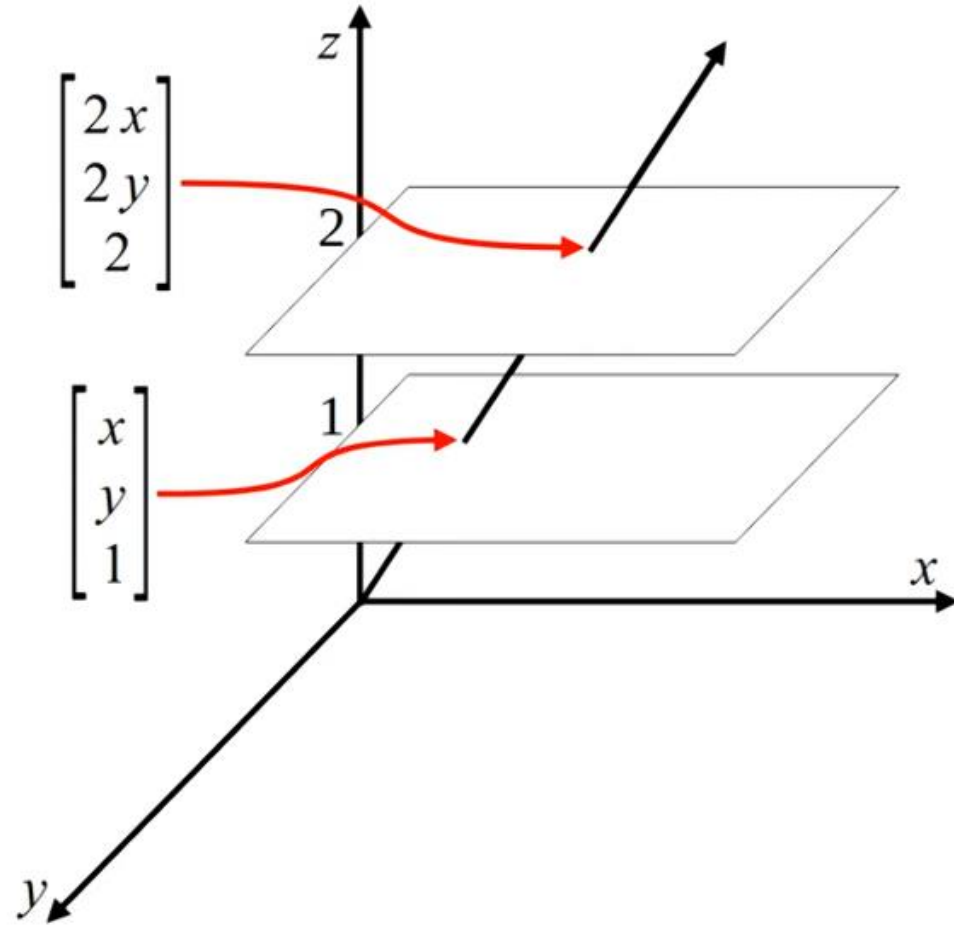
To obtain homogeneous coordinates from Cartesian coordinates,  
one more dimension is added

$$\begin{bmatrix} x \\ y \end{bmatrix}$$

Cartesian  
coordinates

$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Homogeneous  
coordinates



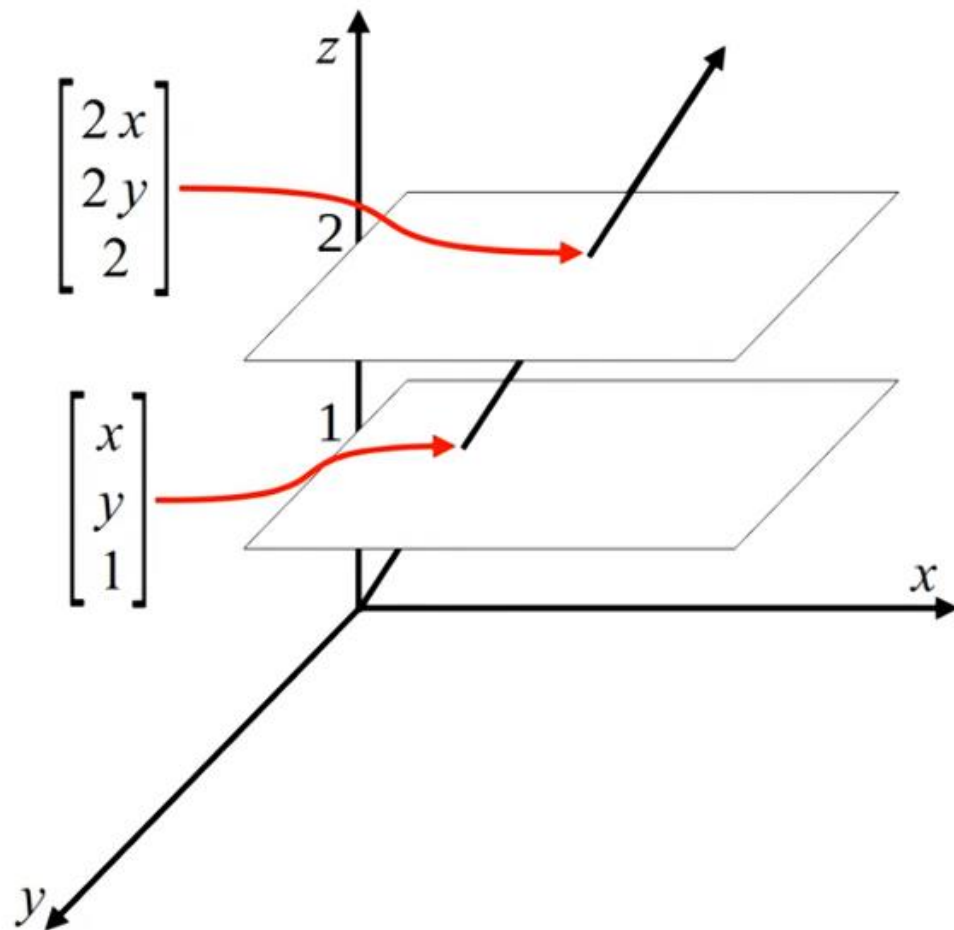
In homogeneous coordinates, a point and its scaled versions are same

$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = w \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} wx \\ wy \\ w \end{bmatrix} \quad w \neq 0$$

They're the same point on different  $z$  planes

$$\begin{bmatrix} x \\ y \\ 0 \end{bmatrix}$$

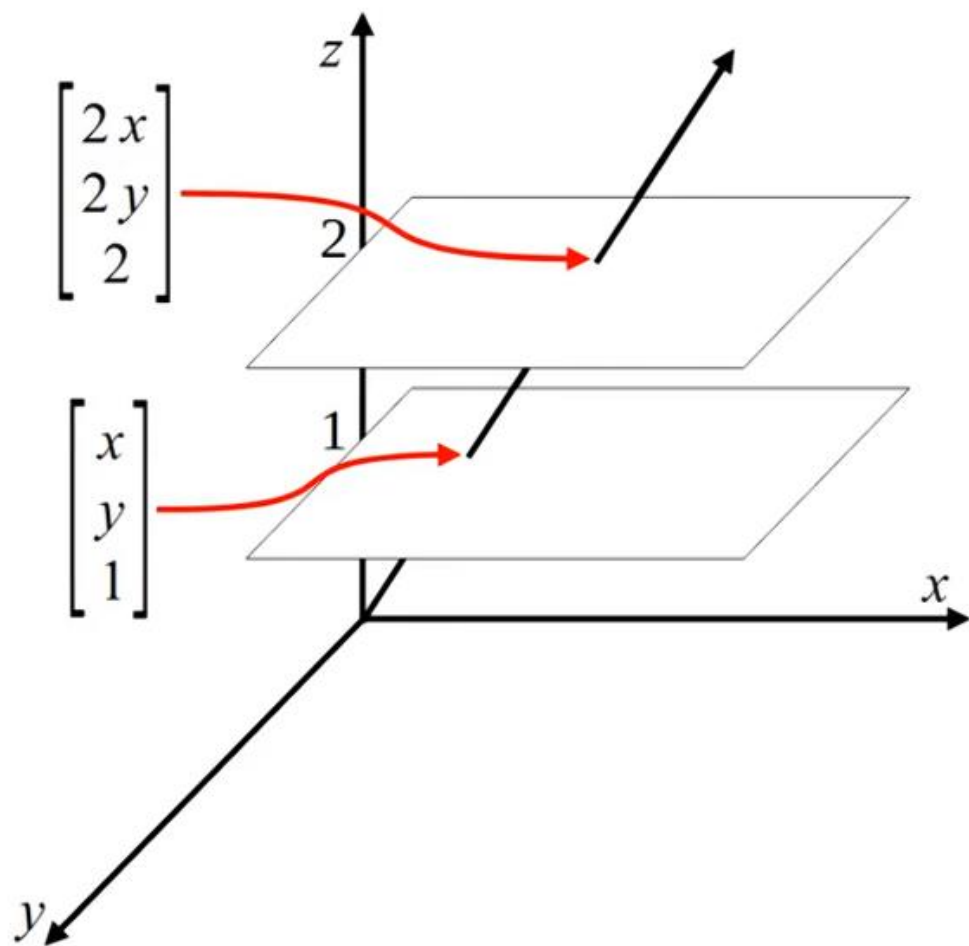
If  $z$  component is zero then the point is at infinity



When going back to Cartesian coordinates

$$\begin{bmatrix} wx \\ wy \\ w \end{bmatrix} \quad \begin{bmatrix} \frac{wx}{w} \\ \frac{wy}{w} \end{bmatrix} \quad w \neq 0$$

Homogeneous coordinates      Cartesian coordinates



When going back to Cartesian coordinates

$$\begin{bmatrix} wx \\ wy \\ w \end{bmatrix} \quad \begin{bmatrix} \frac{wx}{w} \\ \frac{wy}{w} \\ 1 \end{bmatrix} \quad w \neq 0$$

Homogeneous coordinates      Cartesian coordinates

Scaled versions of the same point in homogeneous coordinates correspond to same Cartesian coordinates

# Affine Transformation with Homogeneous Coordinates

Affine Transformation  
in Cartesian coordinates

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} A_{00} & A_{01} \\ A_{10} & A_{11} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} b_0 \\ b_1 \end{bmatrix}$$

One more dimension is added to  
input and output vectors

$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Cartesian  
coordinates

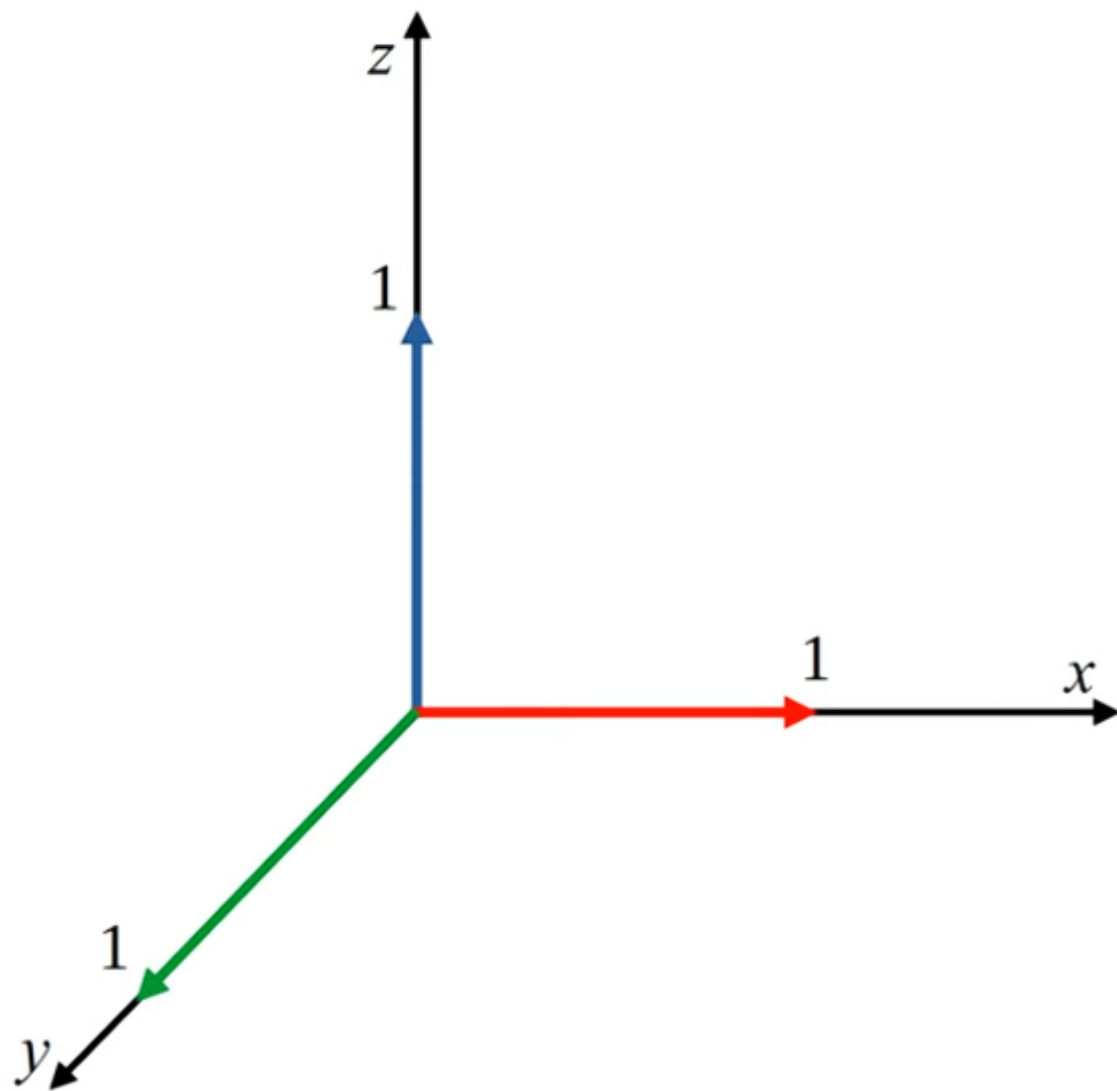
$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix}$$

Homogeneous  
coordinates

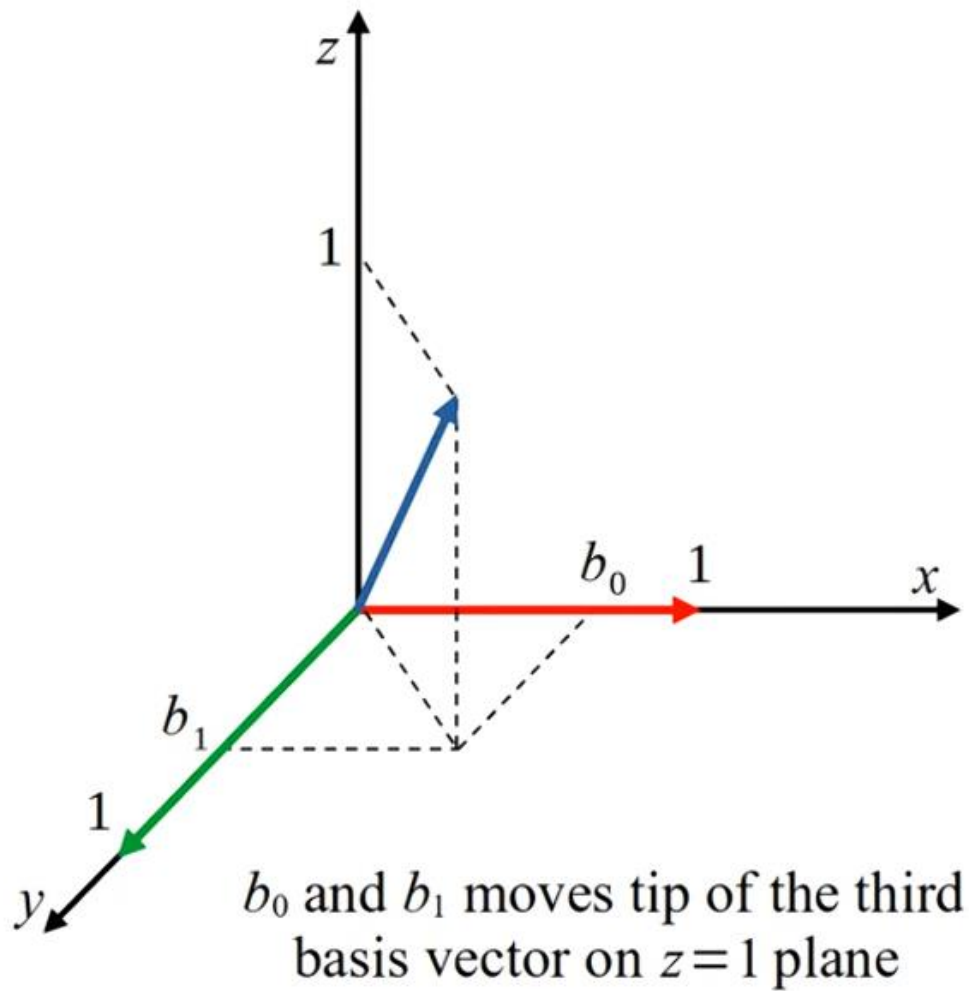
And  $b$  vector is moved inside  
transformation matrix or  
augmented to transformation matrix

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} A_{00} & A_{01} & b_0 \\ A_{10} & A_{11} & b_1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Affine Transformation  
in homogeneous coordinates



Standard basis vectors  
for 3D space



This matrix is a linear transformation matrix in 3D

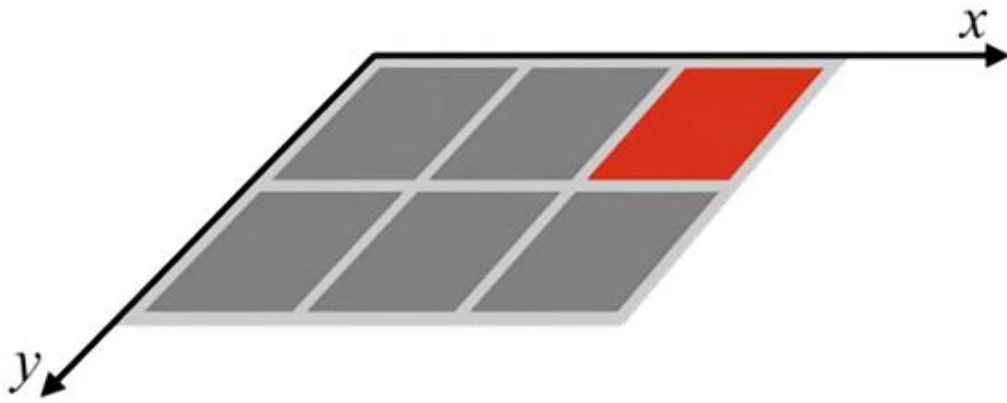
$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} A_{00} & A_{01} & b_0 \\ A_{10} & A_{11} & b_1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

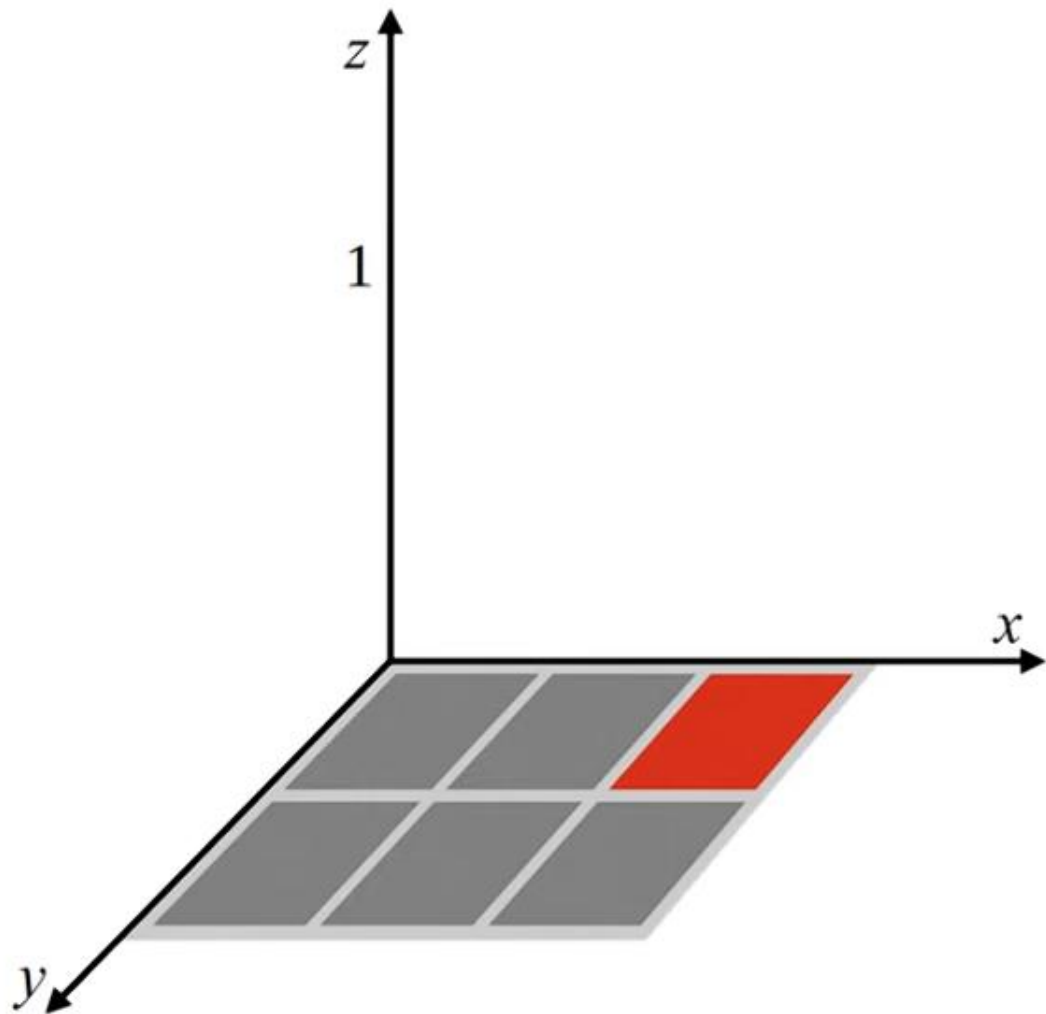
Then this column is the third basis vector of transformed vector space

And what  $b_0$  and  $b_1$  do is to change the orientation of that basis vector

This is our input image in Cartesian coordinates

Image scale in this diagram is not relevant,  
just for illustration

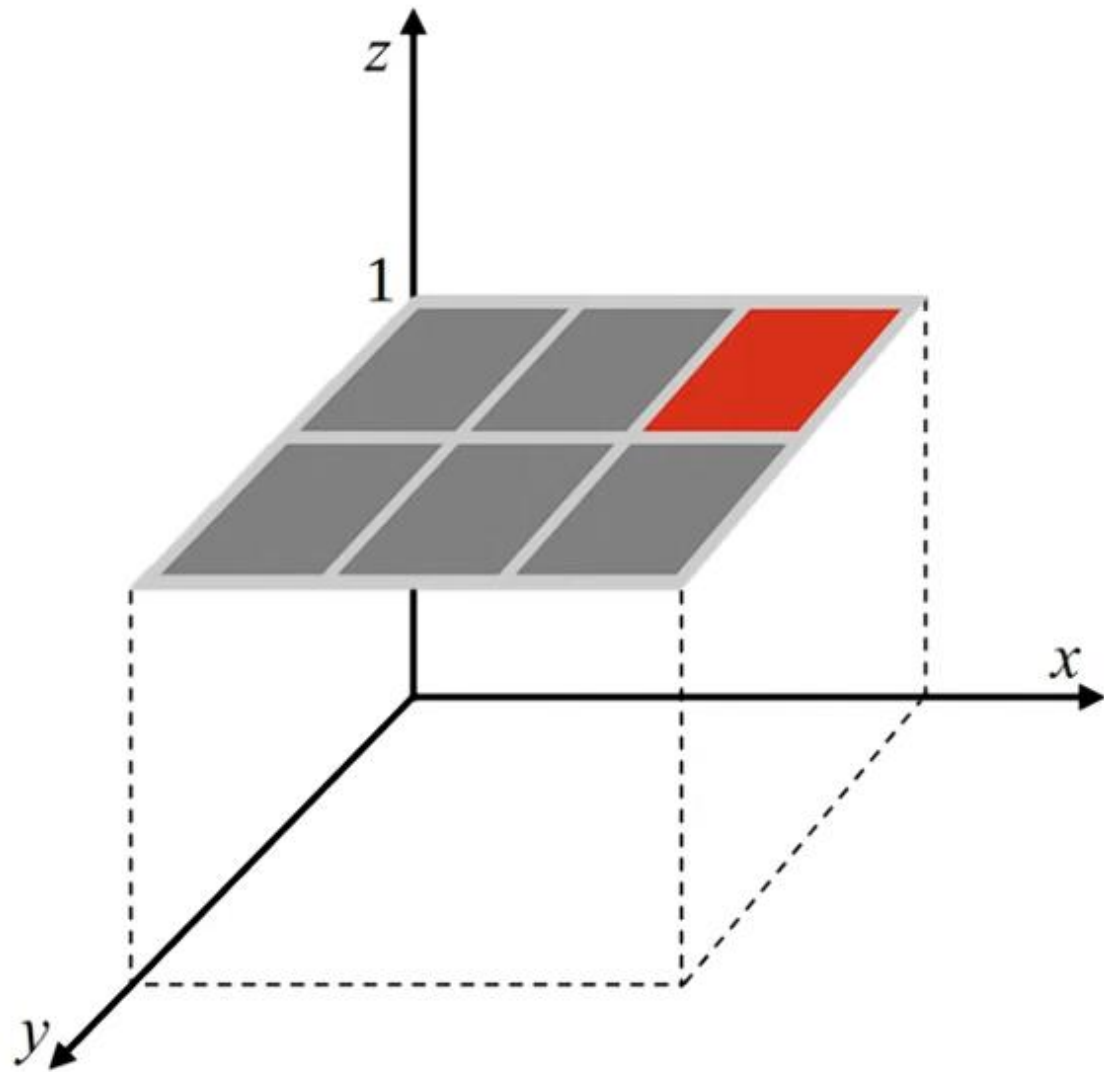




When we move from Cartesian to  
homogeneous coordinates

$$\begin{bmatrix} x \\ y \end{bmatrix} \longrightarrow \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

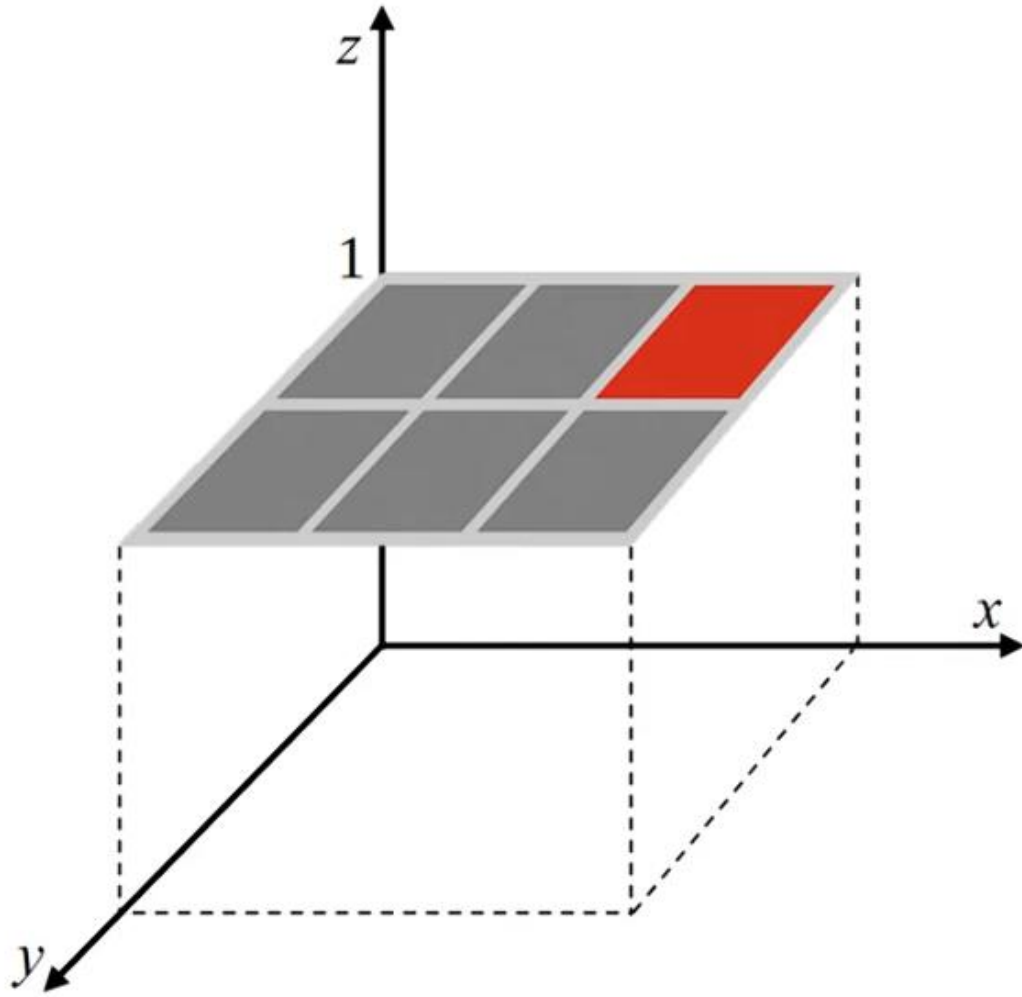
2D image is moved to  $z=1$  plane in 3D



When we move from Cartesian to homogeneous coordinates

$$\begin{bmatrix} x \\ y \end{bmatrix} \longrightarrow \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

2D image is moved to  $z=1$  plane in 3D

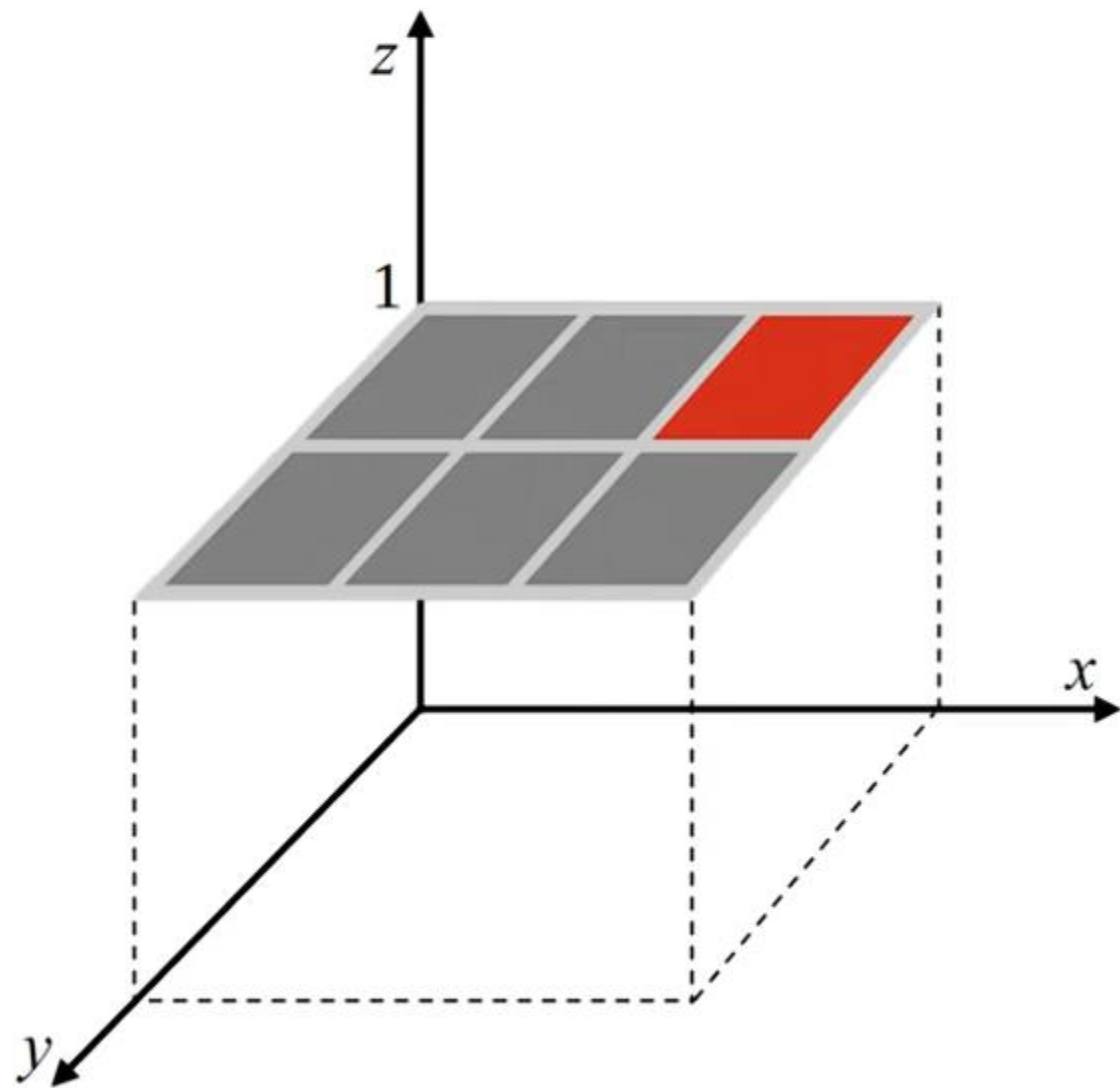


When we move from Cartesian to homogeneous coordinates

$$\begin{bmatrix} x \\ y \end{bmatrix} \longrightarrow \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

2D image is moved to  $z=1$  plane in 3D

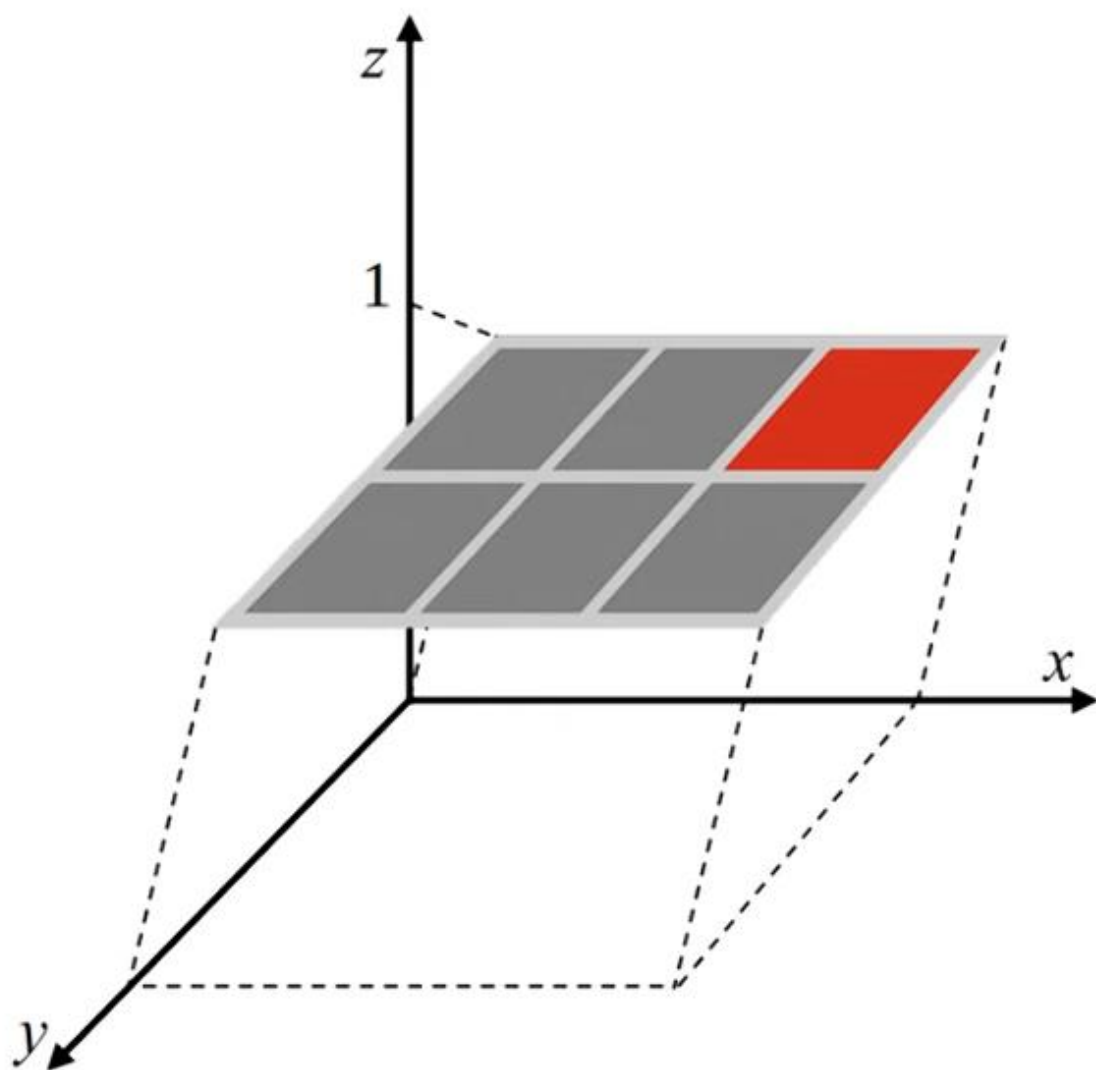
So, 2D image is embedded in 3D



This matrix is a linear transformation matrix in 3D

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} A_{00} & A_{01} & b_0 \\ A_{10} & A_{11} & b_1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$b_0$  and  $b_1$  shear 3D space parallel to  $xy$  plane

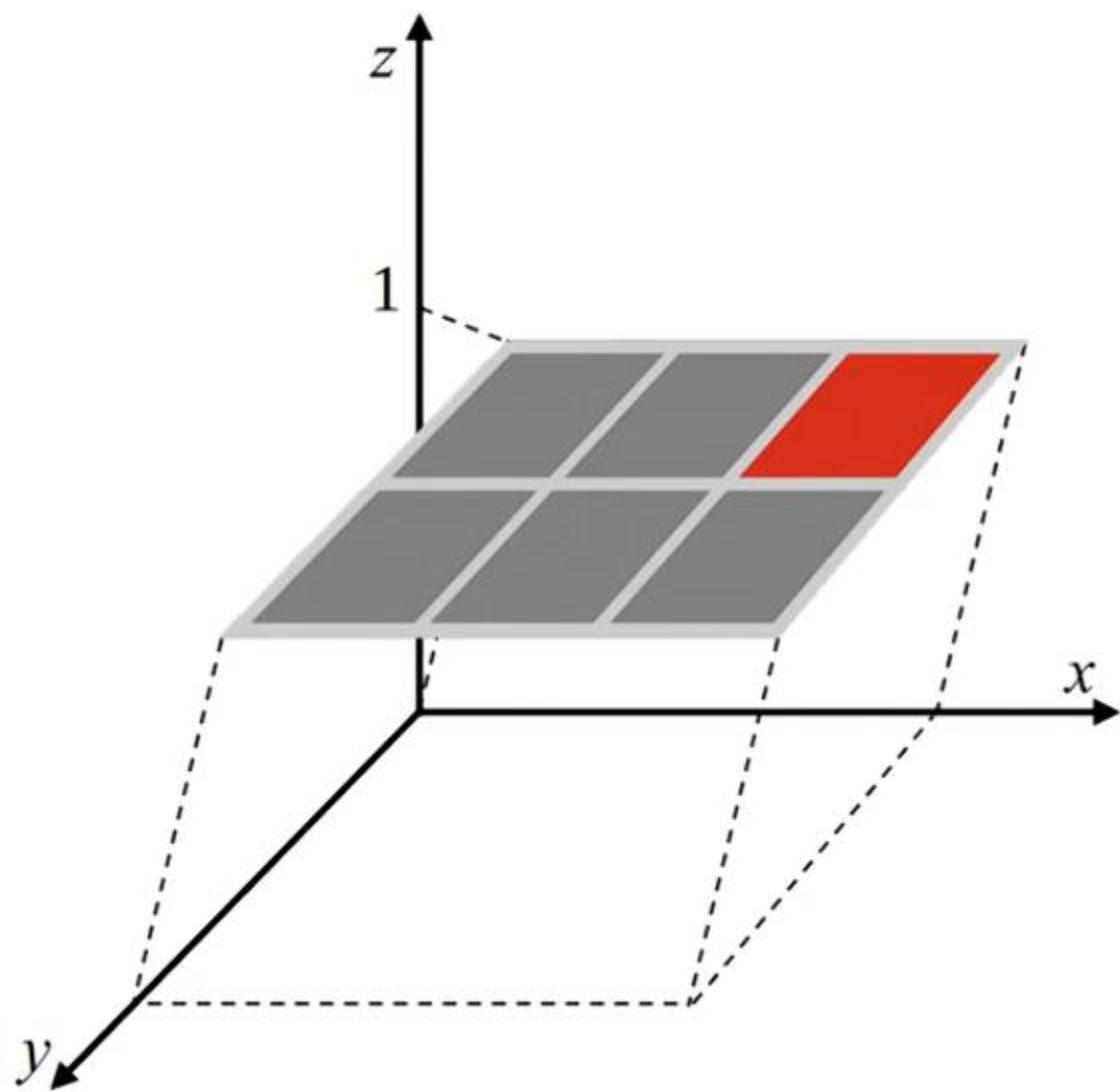


This matrix is a linear transformation matrix in 3D

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} A_{00} & A_{01} & b_0 \\ A_{10} & A_{11} & b_1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$b_0$  and  $b_1$  shear 3D space parallel to  $xy$  plane

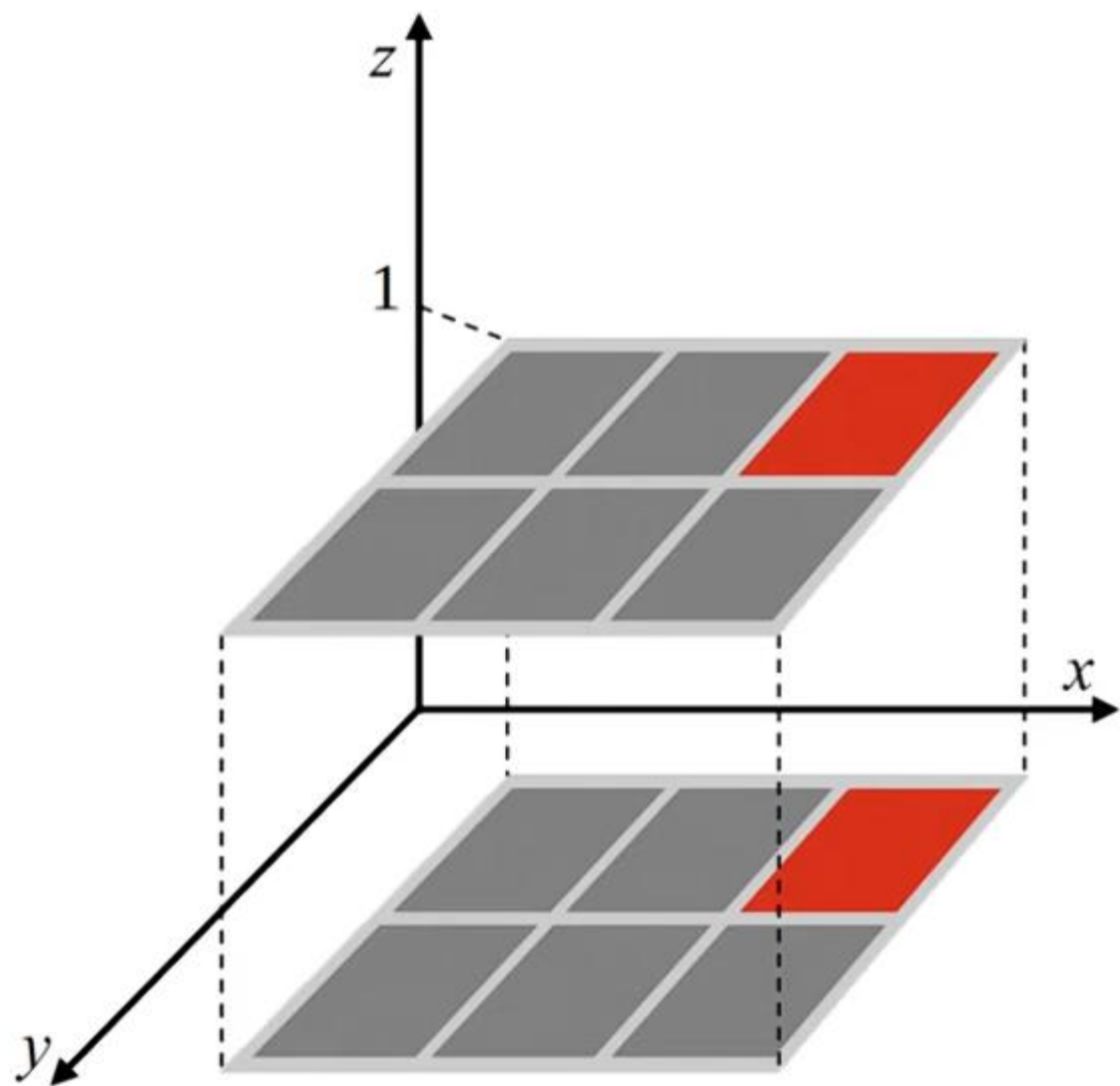
Translation of image over 2D  $z=1$  plane is the result of shear in 3D space



When going back to  
Cartesian coordinates

$$\frac{x'}{z'} = \frac{x'}{1} = x'$$

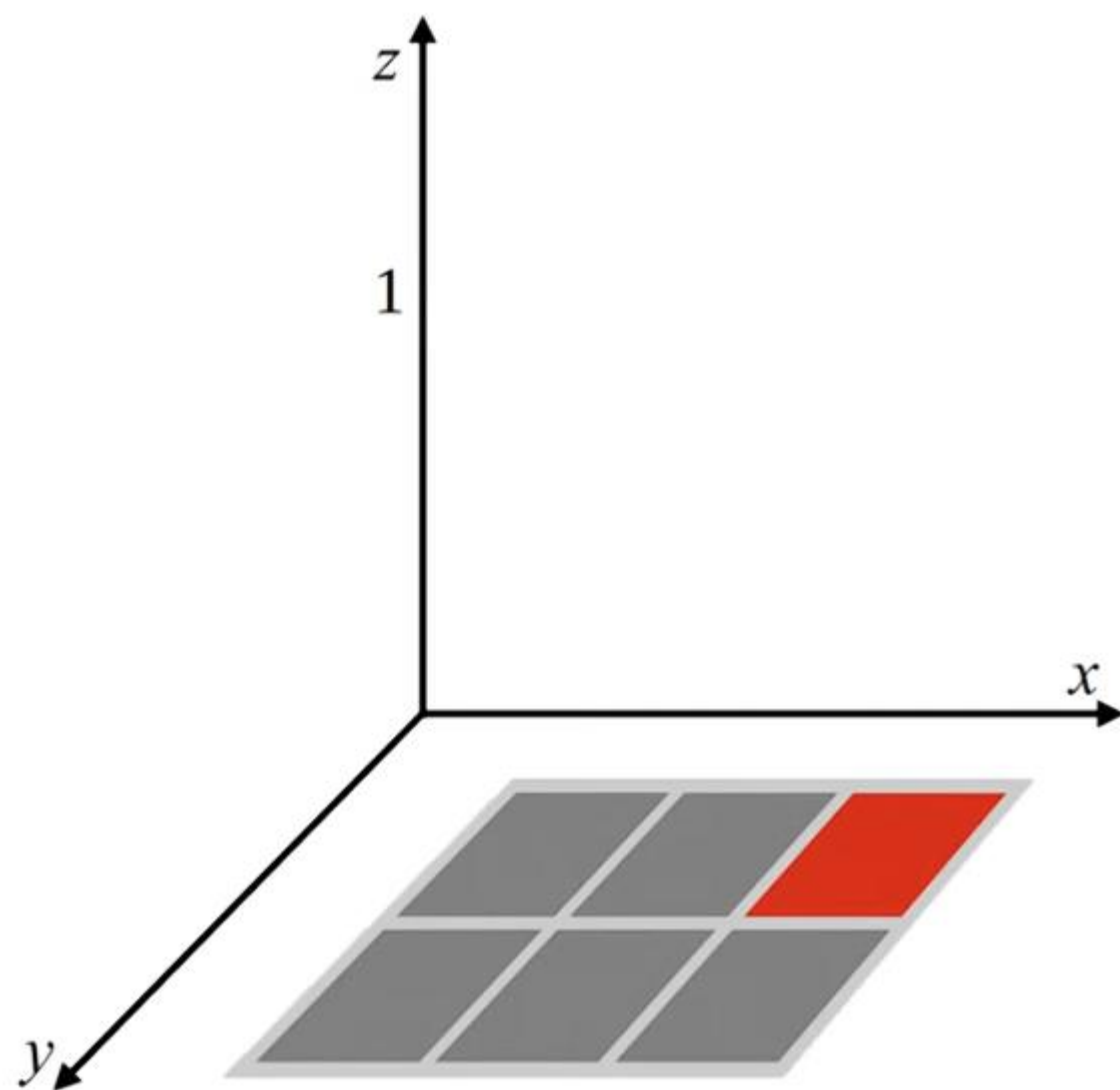
$$\frac{y'}{z'} = \frac{y'}{1} = y'$$



When going back to  
Cartesian coordinates

$$\frac{x'}{z'} = \frac{x'}{1} = x'$$

$$\frac{y'}{z'} = \frac{y'}{1} = y'$$



When going back to  
Cartesian coordinates

$$\frac{x'}{z'} = \frac{x'}{1} = x'$$

$$\frac{y'}{z'} = \frac{y'}{1} = y'$$

# Projective Transformation

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} A_{00} & A_{01} & b_0 \\ A_{10} & A_{11} & b_1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$



$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} A_{00} & A_{01} & b_0 \\ A_{10} & A_{11} & b_1 \\ h_0 & h_1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

What if we change these two entries?  
Let's call them  $h_0$  and  $h_1$

$$h = \begin{bmatrix} h_0 \\ h_1 \end{bmatrix}$$

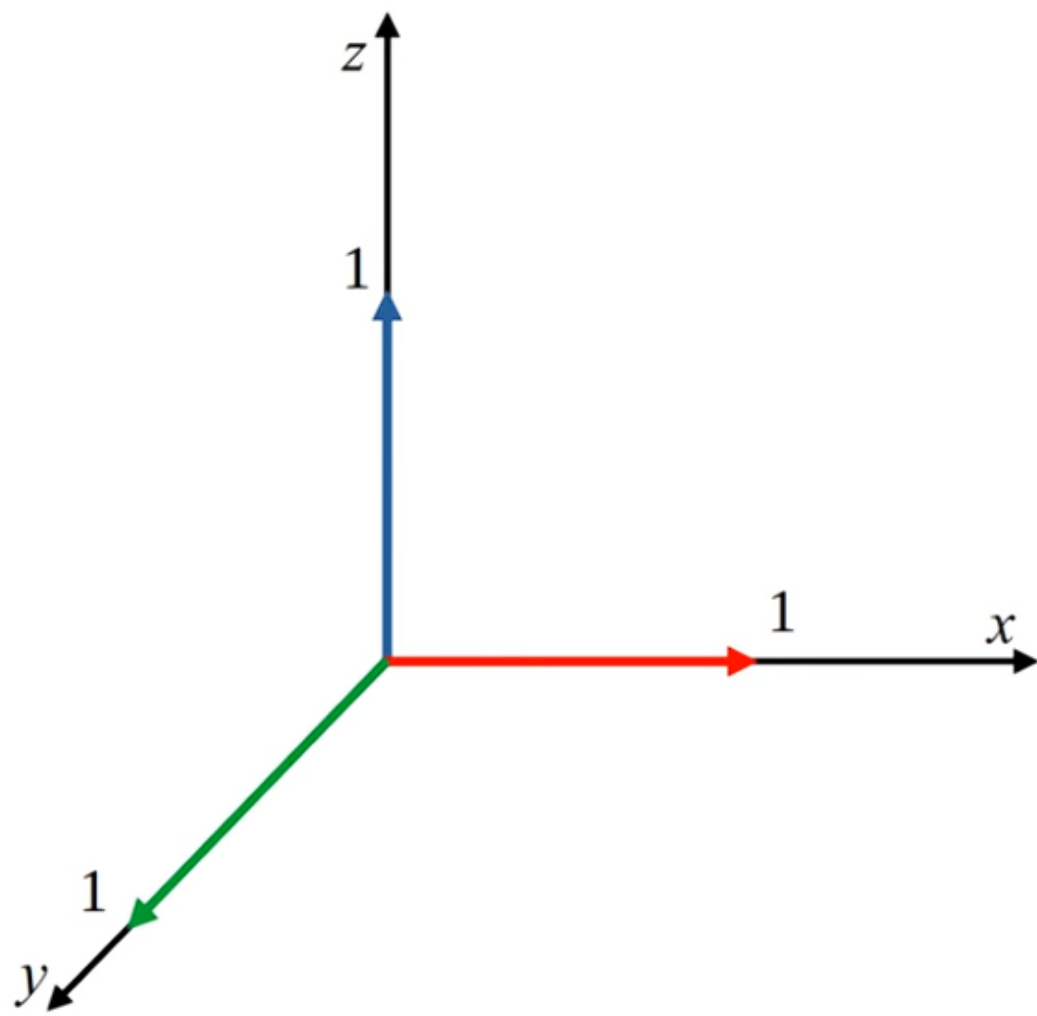
And let  $A$  be identity matrix  
and  $b$  be zero vector

So, we can inspect the  
effect of  $h$  vector

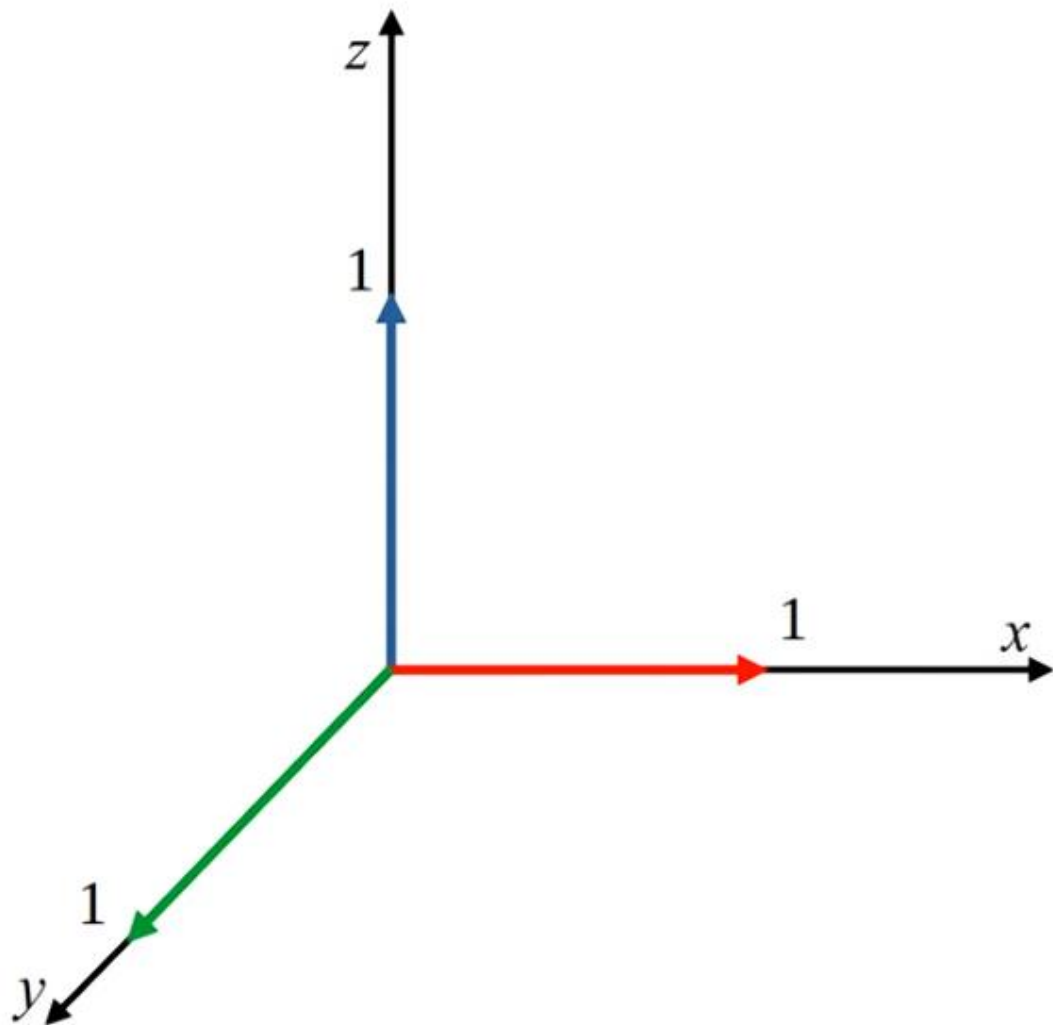


Note that  $z'$  is not 1 anymore because  
 $h$  vector is nonzero

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ h_0 & h_1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$



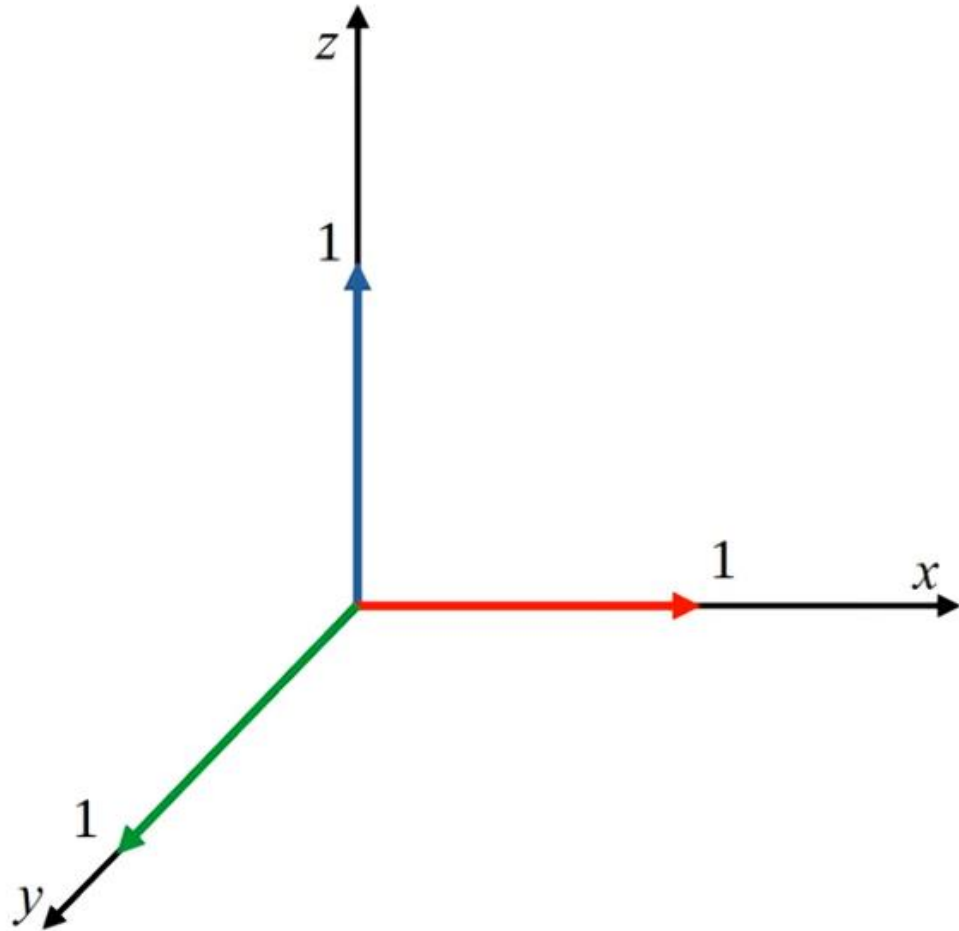
Standard basis vectors  
for 3D space



This matrix is a linear transformation matrix in 3D

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ h_0 & h_1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Then these two columns are the first and the second basis vectors of transformed vector space

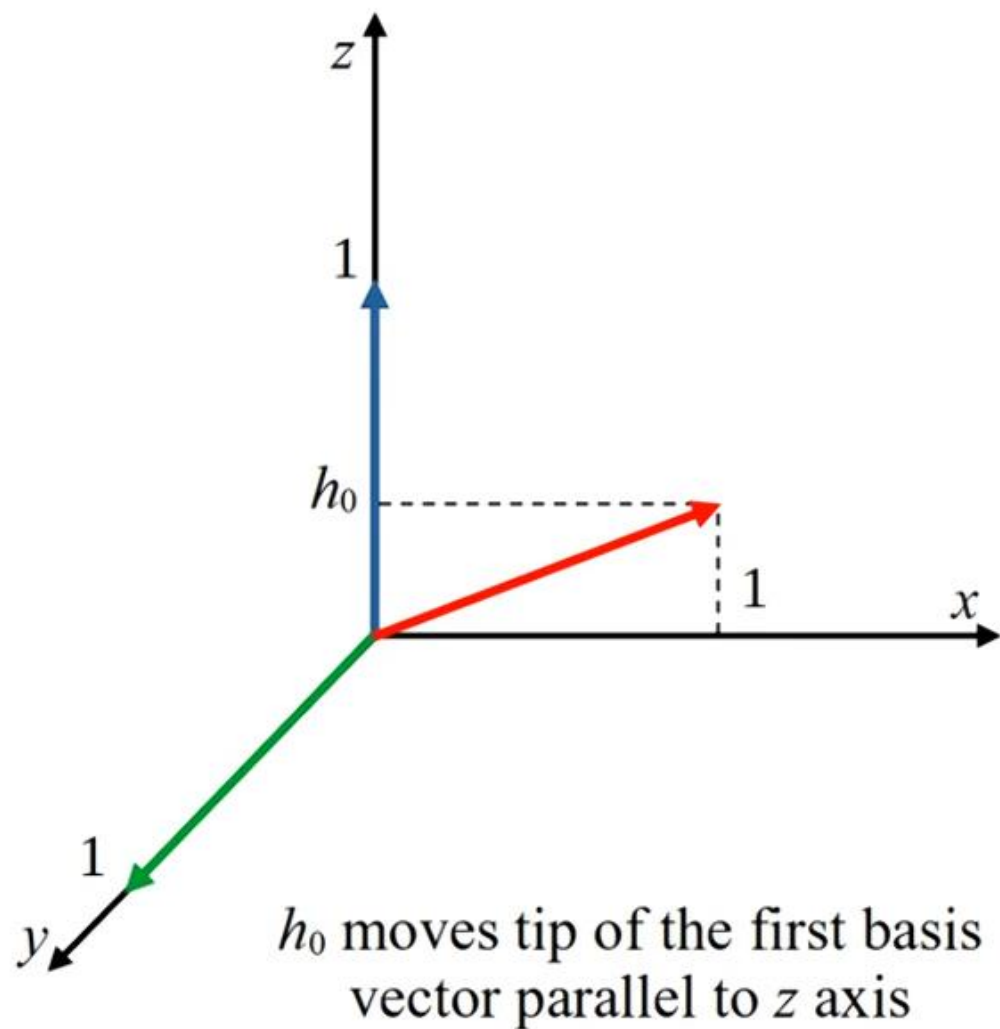


This matrix is a linear transformation matrix in 3D

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ h_0 & h_1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Then these two columns are the first and the second basis vectors of transformed vector space

And what  $h_0$  and  $h_1$  do is to change the orientation of those basis vectors

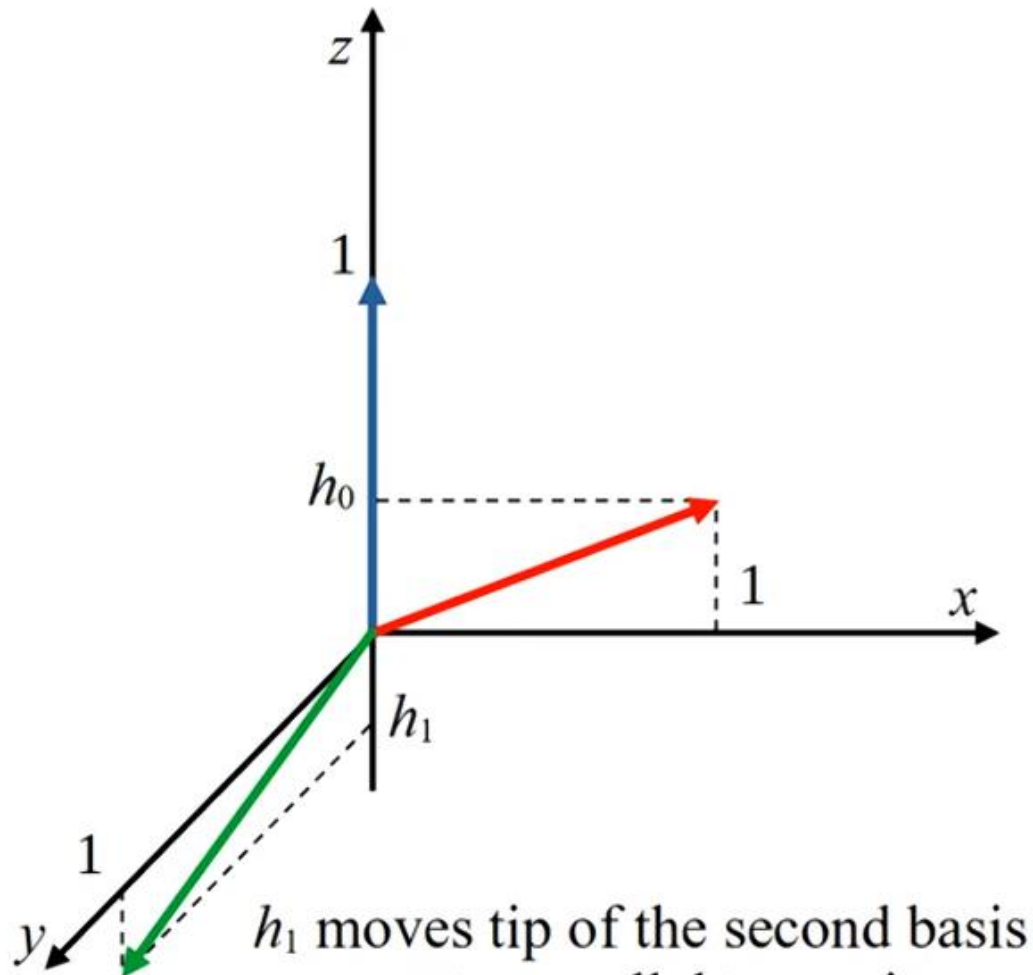


This matrix is a linear transformation matrix in 3D

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ h_0 & h_1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Then these two columns are the first and the second basis vectors of transformed vector space

And what  $h_0$  and  $h_1$  do is to change the orientation of those basis vectors



$h_1$  moves tip of the second basis vector parallel to  $z$  axis

This matrix is a linear transformation matrix in 3D

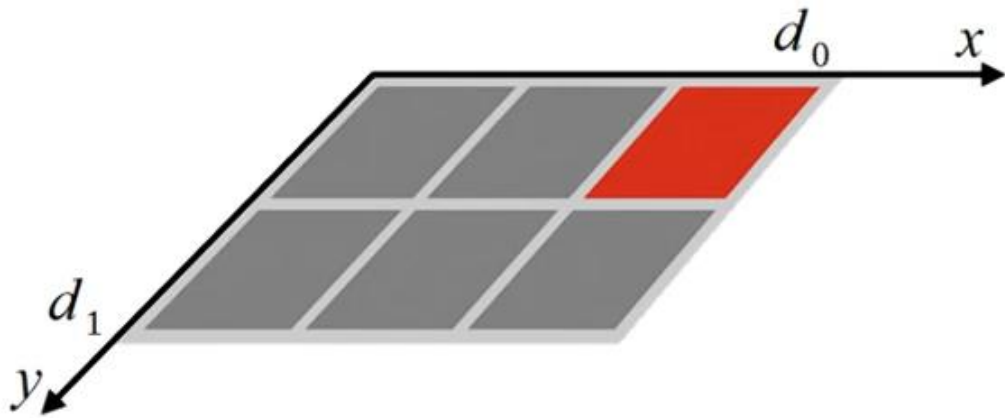
$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ h_0 & h_1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Then these two columns are the first and the second basis vectors of transformed vector space

And what  $h_0$  and  $h_1$  do is to change the orientation of those basis vectors

This is our input image in Cartesian coordinates

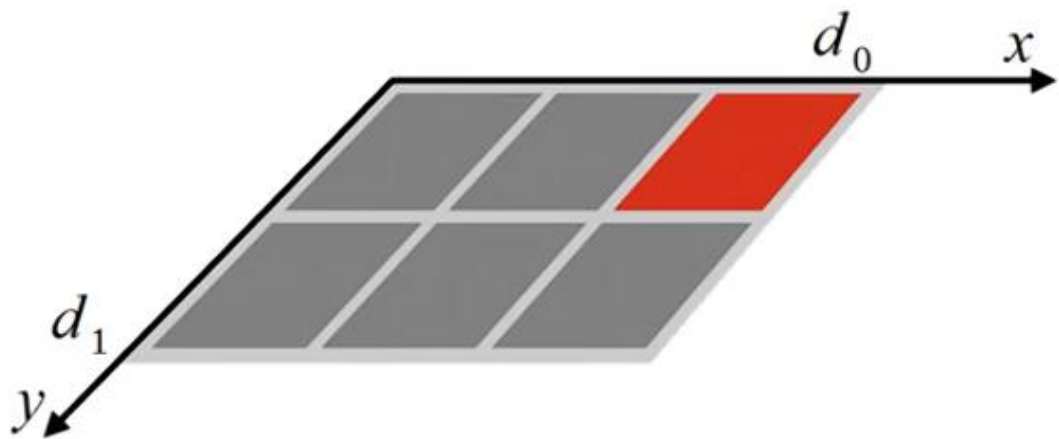
Image scale in this diagram is not relevant,  
just for illustration

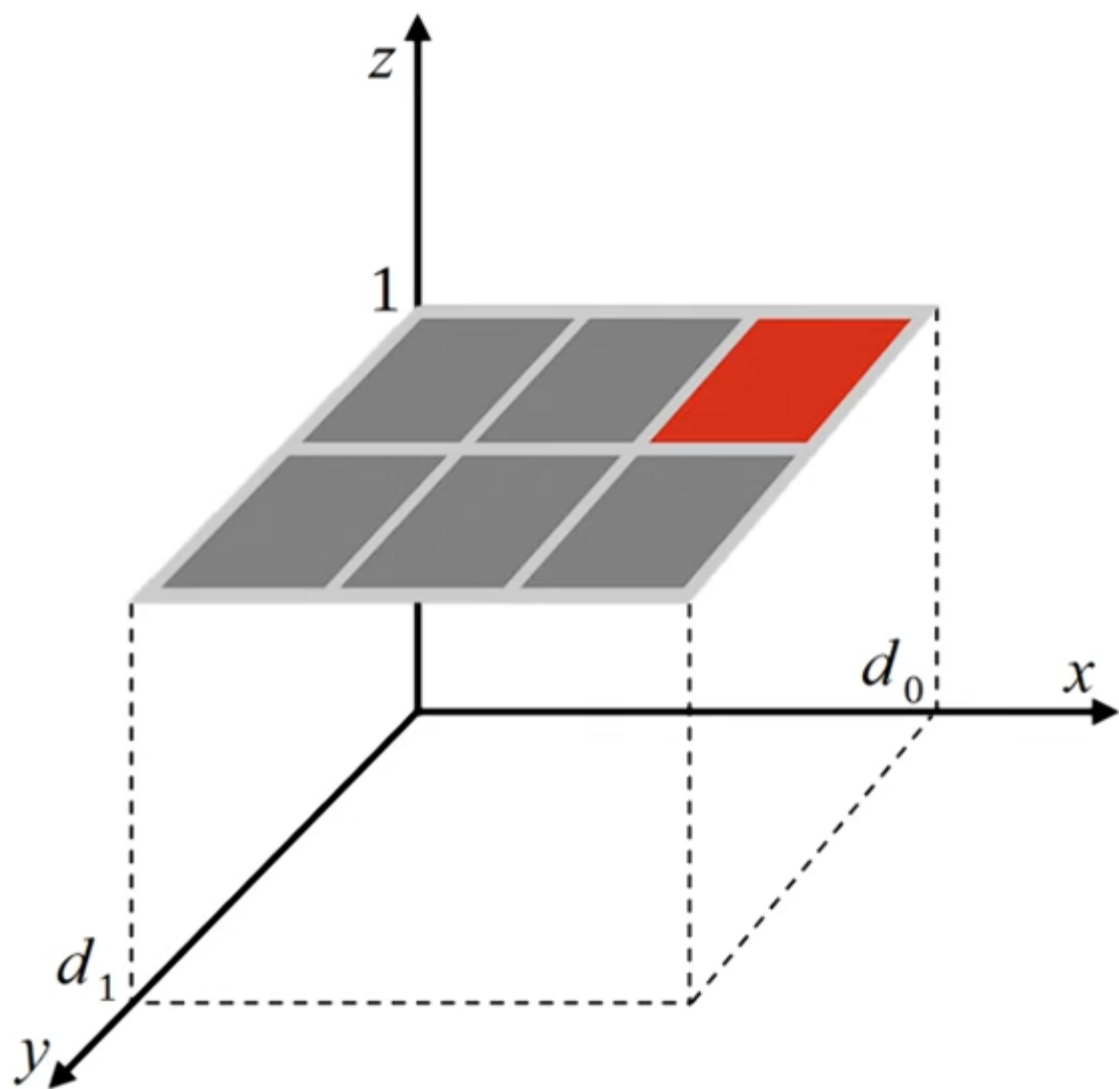


This is our input image in Cartesian coordinates

Image scale in this diagram is not relevant,  
just for illustration

$d_0$  and  $d_1$  are image dimensions,  
width and height (here  $d_0$  equals  $d_1$ )

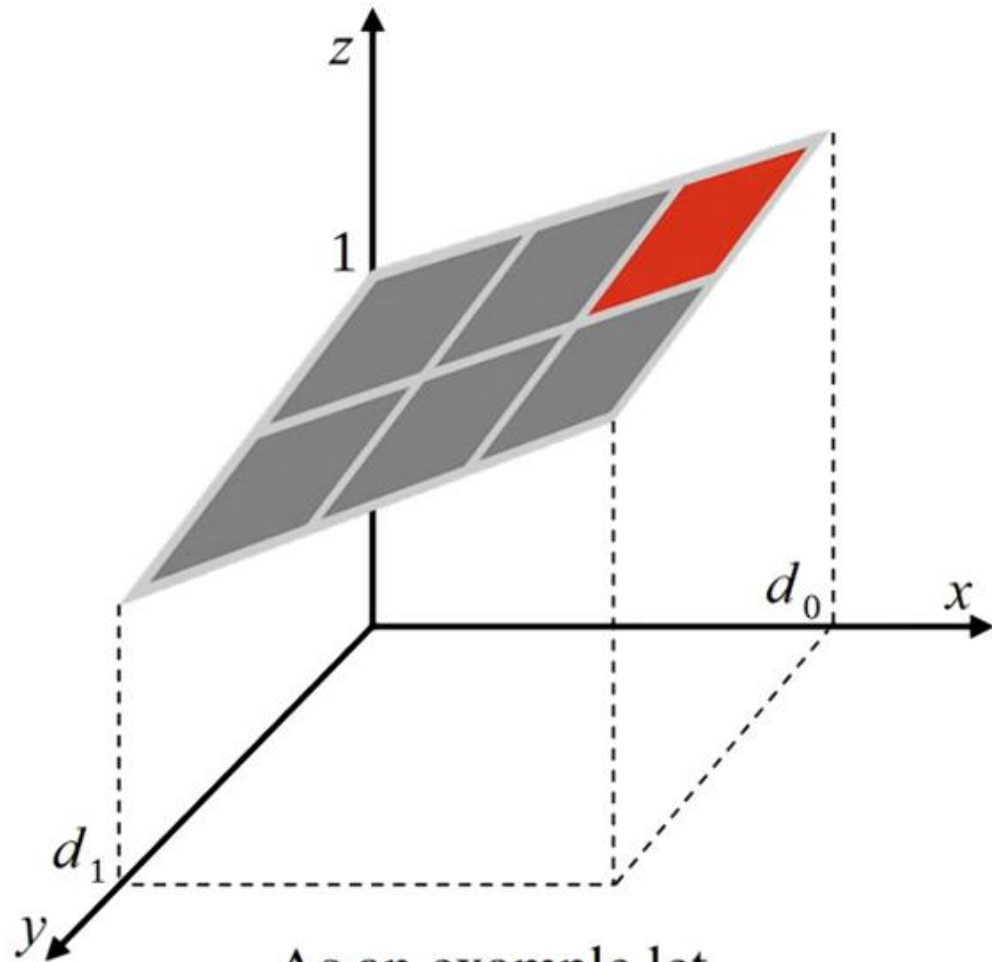




When we move from Cartesian to homogeneous coordinates

$$\begin{bmatrix} x \\ y \end{bmatrix} \longrightarrow \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

2D image is moved to  $z=1$  plane in 3D



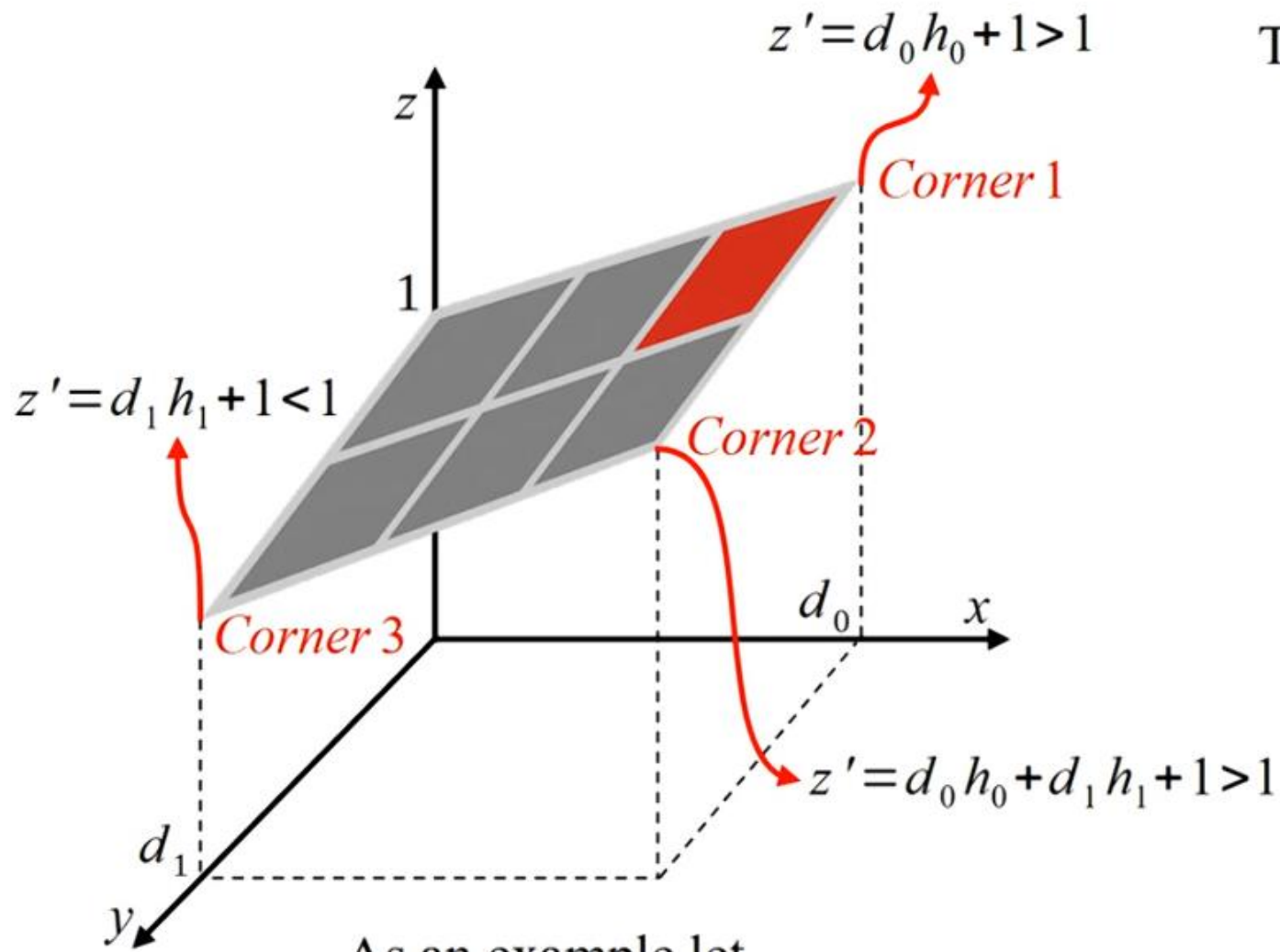
As an example let  
 $h_0 > 0$ ,  $h_1 < 0$  and  $|h_0| > |h_1|$

This matrix is a linear transformation  
 matrix in 3D

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ h_0 & h_1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$h_0$  and  $h_1$  shear 3D space  
 parallel to  $z$  axis

$$z' = h_0 x + h_1 y + 1$$



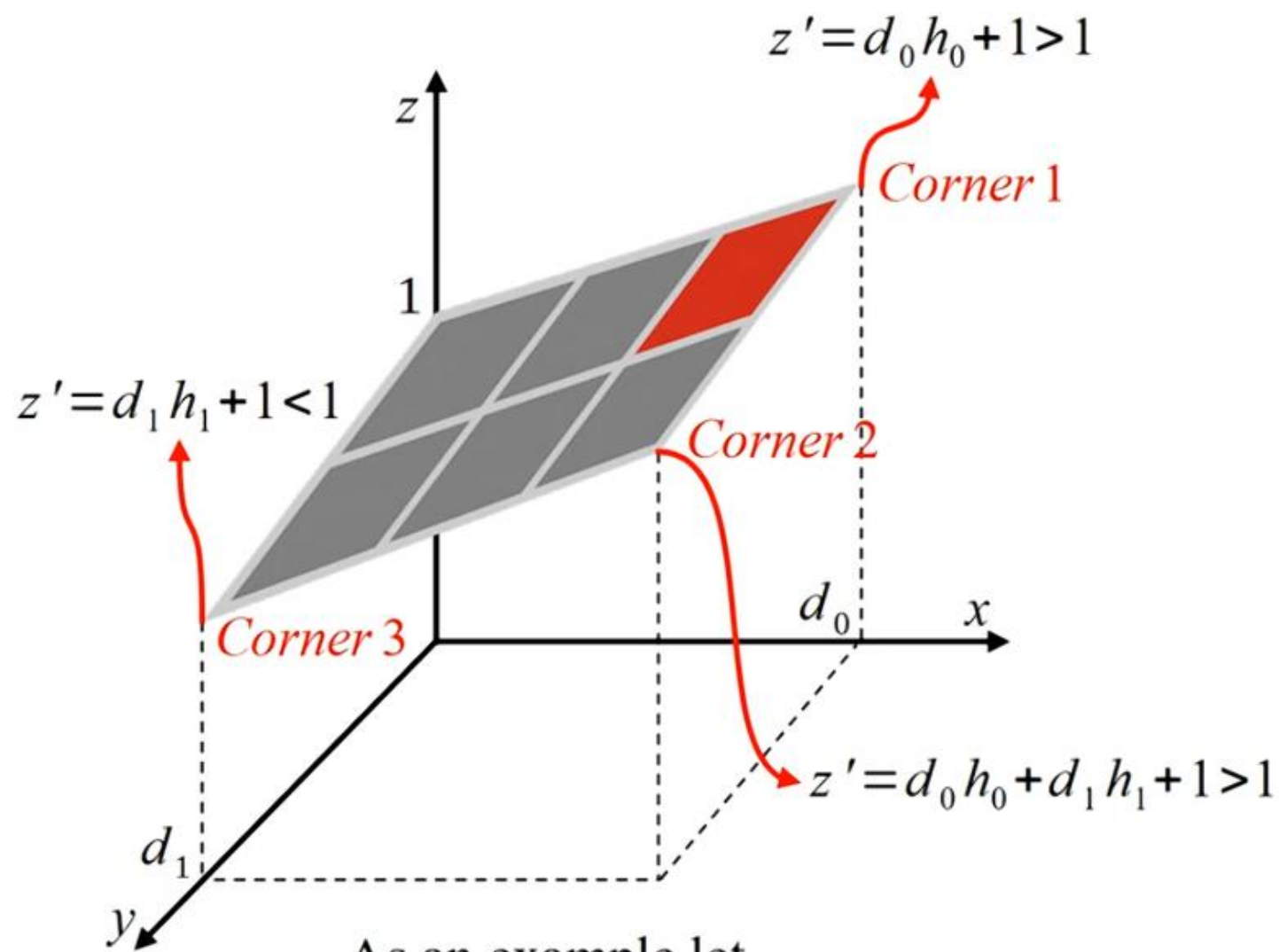
As an example let  
 $h_0 > 0$ ,  $h_1 < 0$  and  $|h_0| > |h_1|$

This matrix is a linear transformation  
 matrix in 3D

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ h_0 & h_1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$h_0$  and  $h_1$  shear 3D space  
 parallel to  $z$  axis

$$z' = h_0 x + h_1 y + 1$$

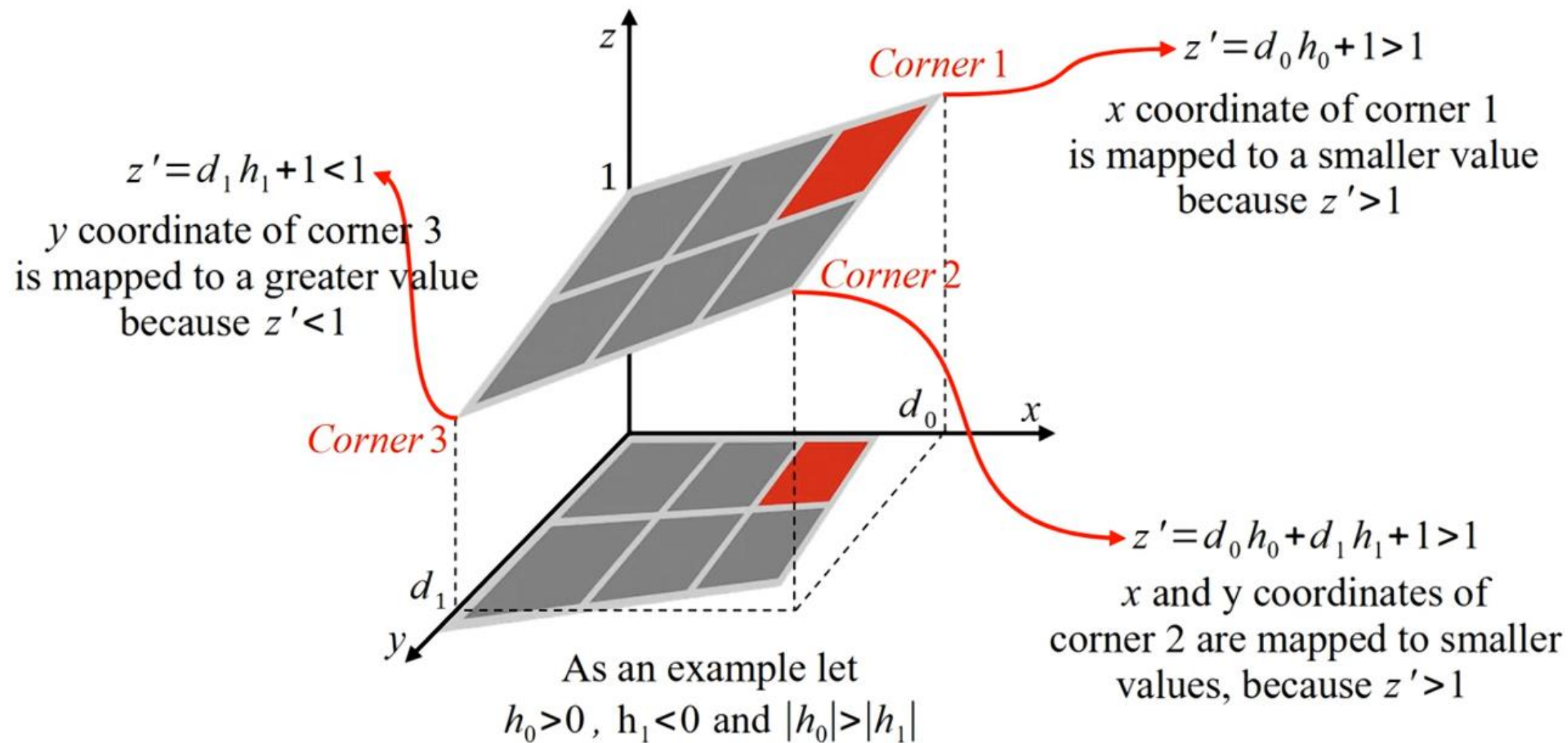


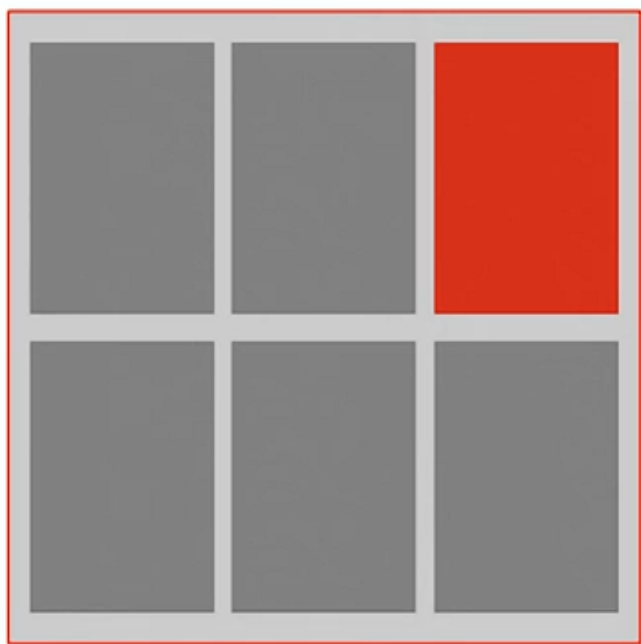
As an example let  
 $h_0 > 0$ ,  $h_1 < 0$  and  $|h_0| > |h_1|$

When going back to  
 Cartesian coordinates

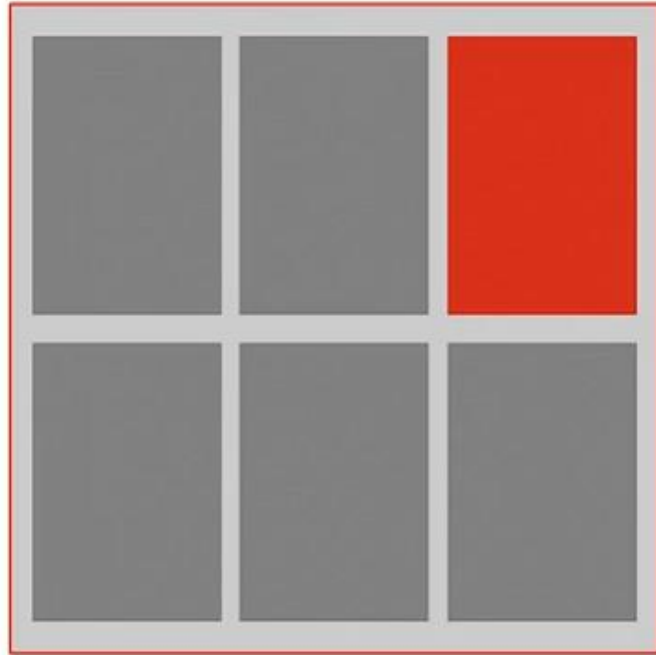
$$\frac{x'}{z'}$$

$$\frac{y'}{z'}$$





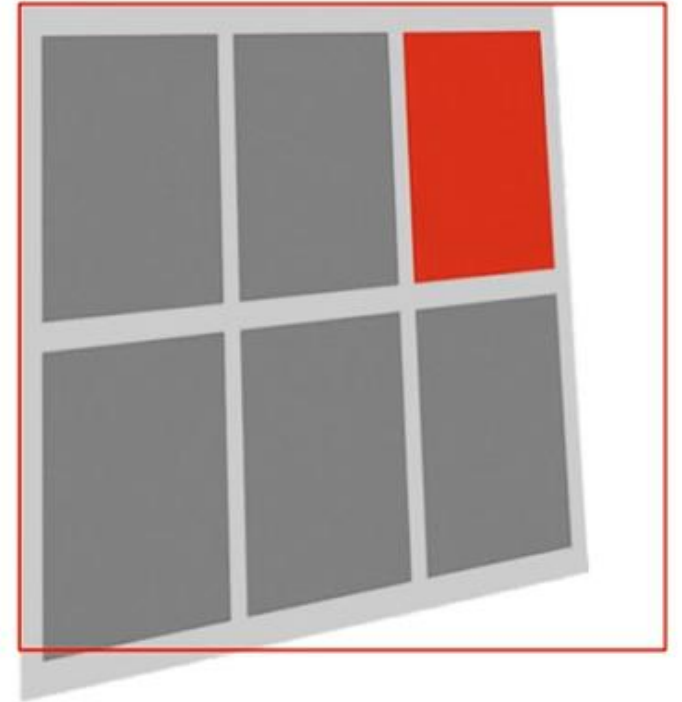
Original Image



Original Image

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ h_0 & h_1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

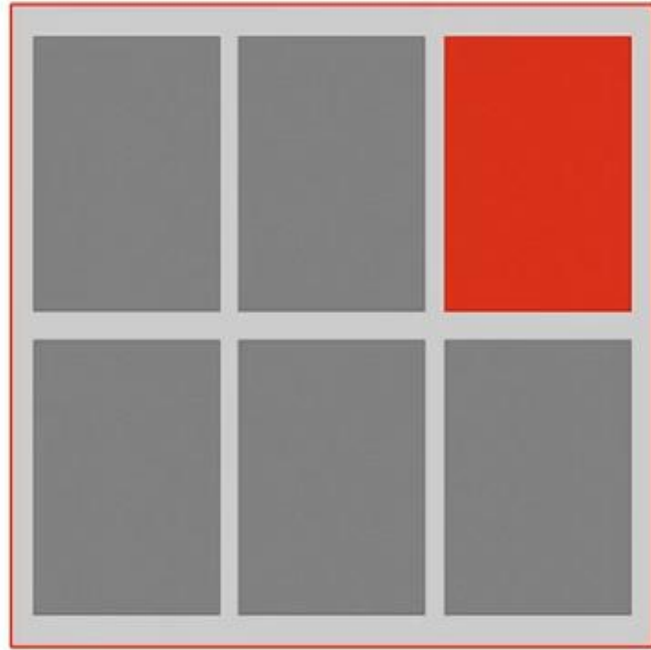
$$h_0 > 0, h_1 < 0 \text{ and } |h_0| > |h_1|$$



Warped Image

Warped image is obtained shearing in 3D space parallel to z axis  
and then mapping result back to 2D Cartesian coordinates

It looks as if the photo is taken from a different perspective

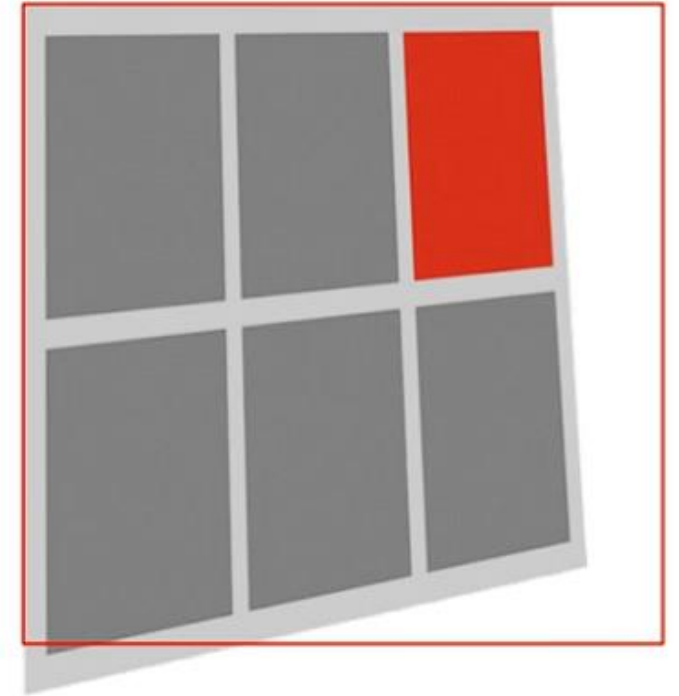


Original Image

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} \boxed{1} & \boxed{0} \\ \boxed{0} & \boxed{1} \\ h_0 & h_1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

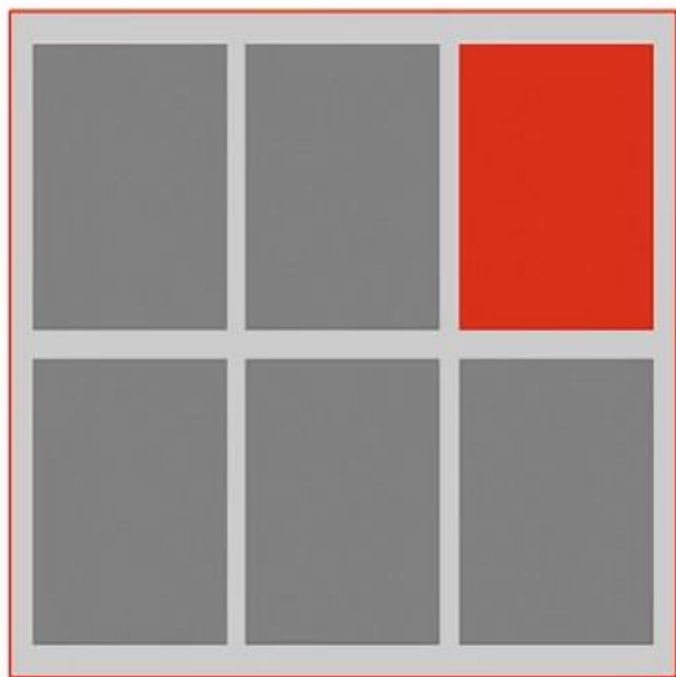
$A$                        $b$

$h_0 > 0, h_1 < 0$  and  $|h_0| > |h_1|$



Warped Image

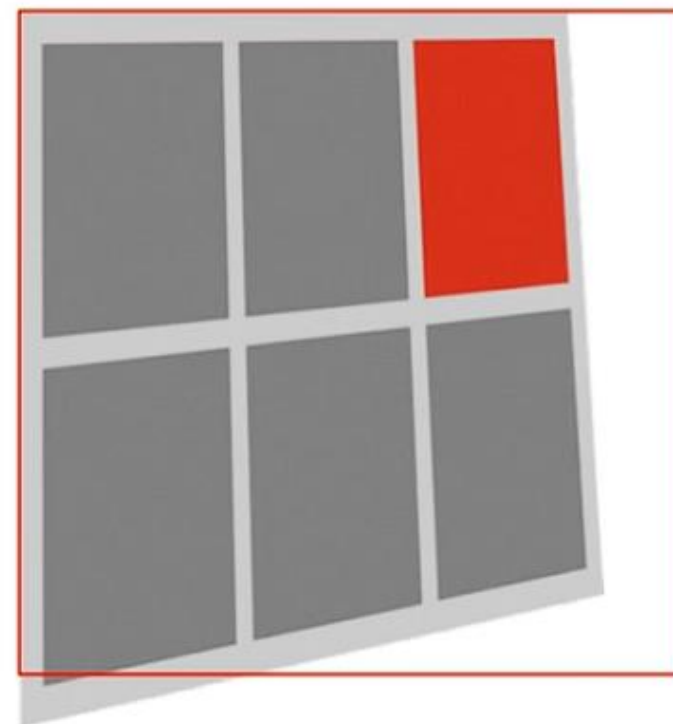
Combining this effect with  $A$  transformation matrix and  $b$  translation vector, different perspectives can be obtained



Original Image

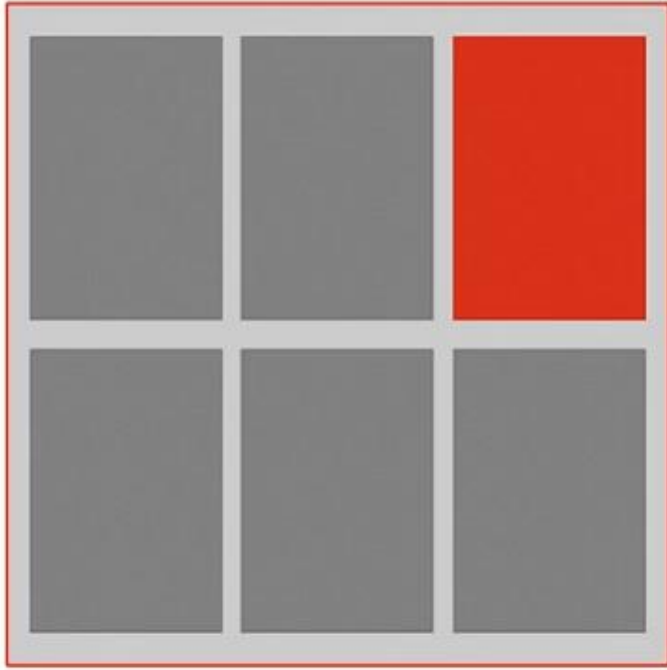
$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ h_0 & h_1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$h_0 > 0, h_1 < 0$  and  $|h_0| > |h_1|$



Warped Image

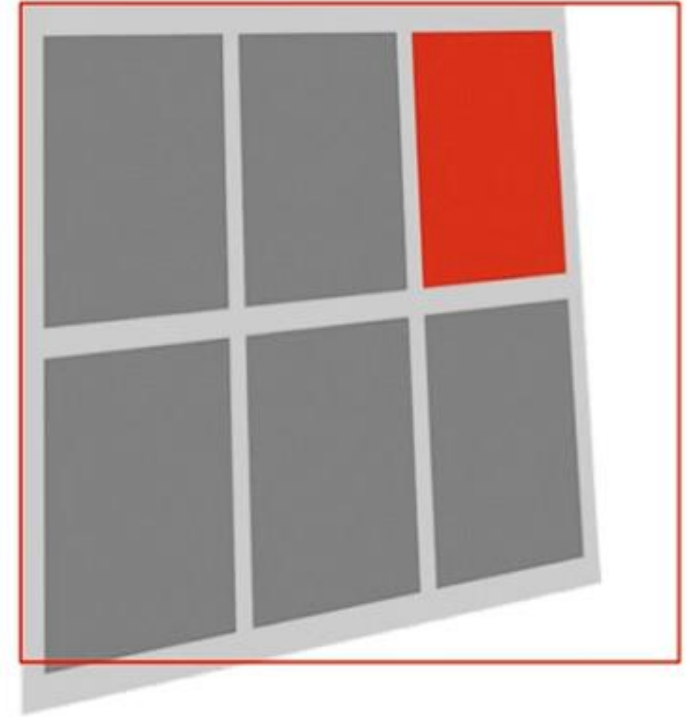
Projective transformation is controlled with 8 parameters



Original Image

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} \overset{A}{\boxed{1 \quad 0}} & \overset{b}{\boxed{0}} \\ \boxed{0 \quad 1} & \boxed{0} \\ h_0 & h_1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$h_0 > 0$ ,  $h_1 < 0$  and  $|h_0| > |h_1|$



Warped Image

Projective transformation has 8 Degrees of Freedom (DoF)

# Reference

- <https://www.youtube.com/watch?v=S7zli1qWKfo>